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Toxicity of bisphenol A alternatives: A study on bisphenol E in two generations of the freshwater snail *Lymnaea stagnalis*

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ABSTRACT

Bisphenol E (BPE), a substitute for bisphenol A (BPA), is becoming increasingly prevalent in the market and, consequently, the environment. Yet, the effects of such alternatives on aquatic organisms remain largely unknown. This study focuses on assessing the developmental toxicity of BPE in embryos (1.1 – 11,946 μg/L) from unexposed parents, as well as its reprotoxicity in adults (4.0 – 872 μg/L) of the Great Pond Snail *Lymnaea stagnalis*. We also examined the effects of BPE on F1 generation embryos (6.6 – 1037 μg/L) from exposed parents. Our study revealed developmental toxicity in embryos from unexposed parents from 5151 μg/L BPE, where the length, the heart rate, and hatching rate were lower than the control group. The embryo rotation was only lower at 11,946 μg/L. Malformations and delays in advancing through developmental stages were also observed. A reproduction test showed no changes in the number of laid egg masses, growth, and weight gain in snails exposed to up to 873 μg/L BPE. Only survival was affected at 187 and 873 μg/L BPE. Exposure of parents did not affect the development of embryos transferred to clean water but lowered the LOEC to 218 μg/L for F1 embryos kept exposed to BPE, compared to 5151 μg/L for exposed embryos from unexposed parents. BPE quantification using UHPLC-MS/MS revealed significant concentration variations, exceeding 20 % of the nominal concentrations, highlighting the importance of chemical analysis in chronic studies. The increased sensitivity of F1 embryos to BPE emphasizes the need for multigenerational approaches in refining ecotoxicological risk assessments of BPA alternatives.

1. Introduction

Bisphenol A (BPA) is the most produced bisphenol worldwide, with an annual production of 10 million tons (Hyun and Ka, 2024). However, BPA is also a documented endocrine-disrupting chemical (EDC) in vertebrates such as fish (Frenzilli et al., 2021). Legislative restrictions on the use of bisphenol A (BPA) in Europe, Canada, and the USA created the need for BPA analogues. Bisphenol E (BPE) is one of 20 analogues currently used in commercial products (Yang and Yu, 2025). This bisphenol is therefore increasingly found in the environment (Liao et al., 2024). The structural similarities between BPE and BPA suggest they may have similar endocrine-disrupting (ED) effects as those reported with BPA (Rosenmai et al., 2014; Chen et al., 2016; Zhang et al., 2019; Hyun and Ka, 2024). The European Partnership for the Assessment and the Regulation of Chemicals (PARC), launched in 2022, aims to enhance chemical hazard and risk assessment in Europe. One of the priorities is to close data gaps on the effects of BPA alternatives on the environment, among which is BPE (PARC, 2024).

The presence of BPE in aquatic environments raises concerns about its potential adverse effects. Contamination of freshwater has been reported in Europe at concentrations up to 2.77 μg/L, and even in source and drinking water in China (Zhang et al., 2019; Russo et al., 2021). An acute toxicity test on zebrafish larvae (*Danio rerio*) revealed a 96-h LC_{50} = 13,610 μg/L for BPE (Gao et al., 2022). Other tests showed a 15-min EC_{50} at 7600 μg/L BPE on the luminescence of the bacterium *Vibrio fischeri*, and a 48-h EC_{50} at 18,000 μg/L BPE on the reproduction of *Daphnia magna* (Chen et al., 2002; Coha et al., 2001). Comparatively, in the larvae of zebrafish *D. rerio*, the 48-h LC_{50} for BPA is 15,170 μg/L BPA (Chen et al., 2012). BPA has EC_{50} values of 2400 μg/L and 10,000 μg/L for *V. fischeri* and *D. magna*, respectively (Chen et al., 2002; Plahuta et al., 2014; Gao et al., 2022). However, research on BPE accounts for <1 % of the studies on BPA alternatives on aquatic vertebrates, and even less on aquatic invertebrates, although invertebrates play a central role in aquatic ecosystems (Bouly et al., 2022; Adamovsky et al., 2024).

Aquatic invertebrates have been used for decades as sentinels for the ecotoxicological status of ecosystems (Sultan and Pei, 2023). The

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Organisation for Economic Co-operation and Development (OECD) has more recently underscored the potential of molluscs in ecotoxicological testing. This effort included a comprehensive review paper and the publication of a reproduction test guideline (TG) for the Great Pond Snail *Lymnaea stagnalis* (OECD, 2010, 2016). This species is easy to maintain and has been shown to respond to various toxicants, including endocrine-disrupting chemicals (EDCs) (Amorim et al., 2019; Gnatyshyna et al., 2023). Yet, given the key differences in the endocrine gene regulation system of molluscs compared to vertebrates, the effects of EDCs in this phylum remain inadequately understood (Scott, 2012; Morthorst et al., 2022). Furthermore, only a few studies investigated the transgenerational impacts of chemicals on mollusc species, including L. *stagnalis* (Reátegui-Zirena et al., 2017; Po and Chiu, 2018; Bouly et al., 2022). Further research on this phylum and BPA alternatives is essential to deepen our understanding of the toxicity mechanisms of EDCs not only in molluscs, but, more generally, invertebrates (Crane et al., 2022).

This study aimed to assess the developmental and reproductive toxicity of BPE on L. *stagnalis* through direct exposure and its impact on the development of F1 generation offspring. Three experiments were conducted, i.e., an embryo test, a reproduction test, and an intergenerational study. We hypothesized that BPE would exhibit comparable effects to other bisphenols, such as BPA, due to their structural similarities. In other invertebrate species, the impact of BPA includes the impairment of embryonic development, such as the disruption of pigment cell differentiation, decreased heart rate, as well as an increase in the reproduction of sexually mature individuals (Schirling et al., 2006; Gomes et al., 2019; Habib et al., 2023).

We aimed to answer three specific questions: 1. Does BPE impair embryonic development in L. *stagnalis*, and if so, to what extent? 2. Does BPE affect the reproduction of L. *stagnalis*? 3. Does parental exposure affect the development of the F1 generation? Our findings will provide new data on the toxicity of BPE and also contribute to establishing the potential of L. *stagnalis* in assessing the developmental and reproductive effects of EDCs in aquatic invertebrate species.

2. Materials and methods

2.1. Test chemical and exposure media

The tap water used in the culturing, breeding, acclimation, and exposure periods was collected after a minimum of 10 min of water flow through the tap to avoid metal contamination from the pipes. BPE (CAS: 2081–08–5) was purchased from Chiron AS (Trondheim, Norway). For exposed embryos, 100 mL of BPE at 51 mg/L was prepared as a stock solution to create 100 mL of exposure solution for each test concentration. For the reproduction test, 400 mL of BPE stock solution of 80 ± 10 mg/L was prepared and diluted into 2 L glass bottles to make exposure media at exposure concentrations. New stock solutions and exposure media were prepared after every second media renewal. For the intergenerational study, the stock solution (80 ± 10 mg/L) from the reproduction test was spiked and diluted to prepare 100 mL of each desired exposure concentration.

2.2. Stability of BPE concentration in the exposure media

The concentration of BPE in exposure water was quantified using a liquid chromatography-tandem mass spectrometry (LC-MS/MS) method on spiked exposure media. The spiked media was directly injected into a UHPLC system (Agilent 1290 Infinity II) without prior sample preparation, using 5 μL, on a reversed-phase C18 column (2.1 × 50 mm, 1.8 μm particle size, Agilent Technologies). The mobile phase consisted of 20 % water and 80 % methanol/acetonitrile (1:1, v/v). The HPLC system was connected to a triple quadrupole system (Agilent 6495C). The data were recorded and processed using the Agilent MassHunter™ software suite.

For exposed embryos, media samples were taken from the stock solutions and the exposure media before renewal three times during the test. For the intergenerational study, media samples were taken from the stock solutions at the start of the test, as well as from the wells before media renewal, four times throughout the test. For the reproduction test, 1.5 mL of water was spiked in all six replicated jars (n = 6 for each concentration group), both before and after media renewal, twice a week. This sampling was carried out until day 24. We calculated the BPE time-weighted average (TWA) concentration, as described in the OECD TG 243, as:

$$TWA = \left[\left[\sum_{i=1}^{n}\left[\frac{(c_i - c_{i+1})(t_{i+1} - t_i)}{ln(c_i) - \ln(c_{i+1})}\right]\right] \bigg/ \left[\sum_{i=1}^{n}[t_{i+1} - t_i]\right]\right]$$

Where c is the measured concentration in μg/L, and t is the time in days.

2.3. Test organism and lab culturing

For our studies, adult L. *stagnalis* were collected from a pond in Klintholm, South-Eastern Funen, Denmark, in fall 2023. This location was chosen for its distance from human installations to avoid on-site snail contamination as much as possible. The snails were kept at the University of Southern Denmark, Odense, in 50 L aquaria in a controlled room, with a constant air temperature of 20 °C, a 16/8 h light/dark cycle, and constant water aeration. After collection from the wild, the snails were kept in pond water for 5 days, followed by an additional 5 days in a mixture of pond and tap water (1:1, v/v). After this acclimatization process, the snails were transferred into 100 % tap water (groundwater). The water in all tanks was completely renewed every week. The snails were fed ad libitum organic lettuce during acclimatization and culturing periods.

2.4. Experiment 1: Embryonic development under direct BPE exposure

An overview of the experimental design is shown in Fig. 1. Before the test, selected tanks in the culturing room were emptied of clutches. On the morning of the test, three newly laid clutches (<24 h old, 80 ± 20 eggs) were gently detached from the glass wall of the aquaria using a blade and placed in a petri dish. Clutches were opened with a scalpel and emptied, and the eggs were gently rolled on filter paper to remove the jelly matter. Eggs were then transferred to tap water and pipetted individually into 96-well plates with 250 μL of tap water. Before exposure, eggs were checked for abnormalities, including empty eggs and multiple embryos. The water was then pipetted out, and 250 μL of exposure solution was added to the assigned wells. The following nominal concentrations were tested as: 0 (control), 1.0, 10, 100, 1000, 5000, and 10,000 μg/L. These concentrations were chosen after an internal range-finding exposure test of 21 days (data not shown). The control group consisted of 28 eggs (n = 28), and each exposure group consisted of 14 eggs (n = 14) at the beginning of the test. The plates were maintained in a controlled room, where they were kept at a constant temperature of 20 °C with a 16/8-hour light/dark cycle. The exposure lasted 21 days, and the water and exposure media were renewed every second day, using the same stock solutions throughout the 21-day test.

A picture of each embryo was taken daily, up to day 10, using a Nikon ECLIPSE Ti Series inverted microscope combined with NIS-Elements imaging software (© 2025 Nikon Instruments Inc). The length of each embryo was measured with ImageJ2 (version 2.9.0/1.53t) on days 1, 2, 3, 5, 7, 9, and 10. The developmental stages were identified as: morula, trochophora, veliger, and hippo (Bandow and Weltje, 2012). In the morula and trochophora stages, the length of the whole embryo was measured. At the veliger and hippo stages, only the length of the shell was measured, from the apex to the basal lip. On day 4, embryonic rotation was quantified with 2-minute videos of each embryo, counting the maximum number of 1/2 rotations, then expressed in revolutions per minute (rpm). On day 6, 1-minute videos of each embryo were taken to quantify the heart rate. Beats were counted in a random 15-second window and then expressed as beats per minute

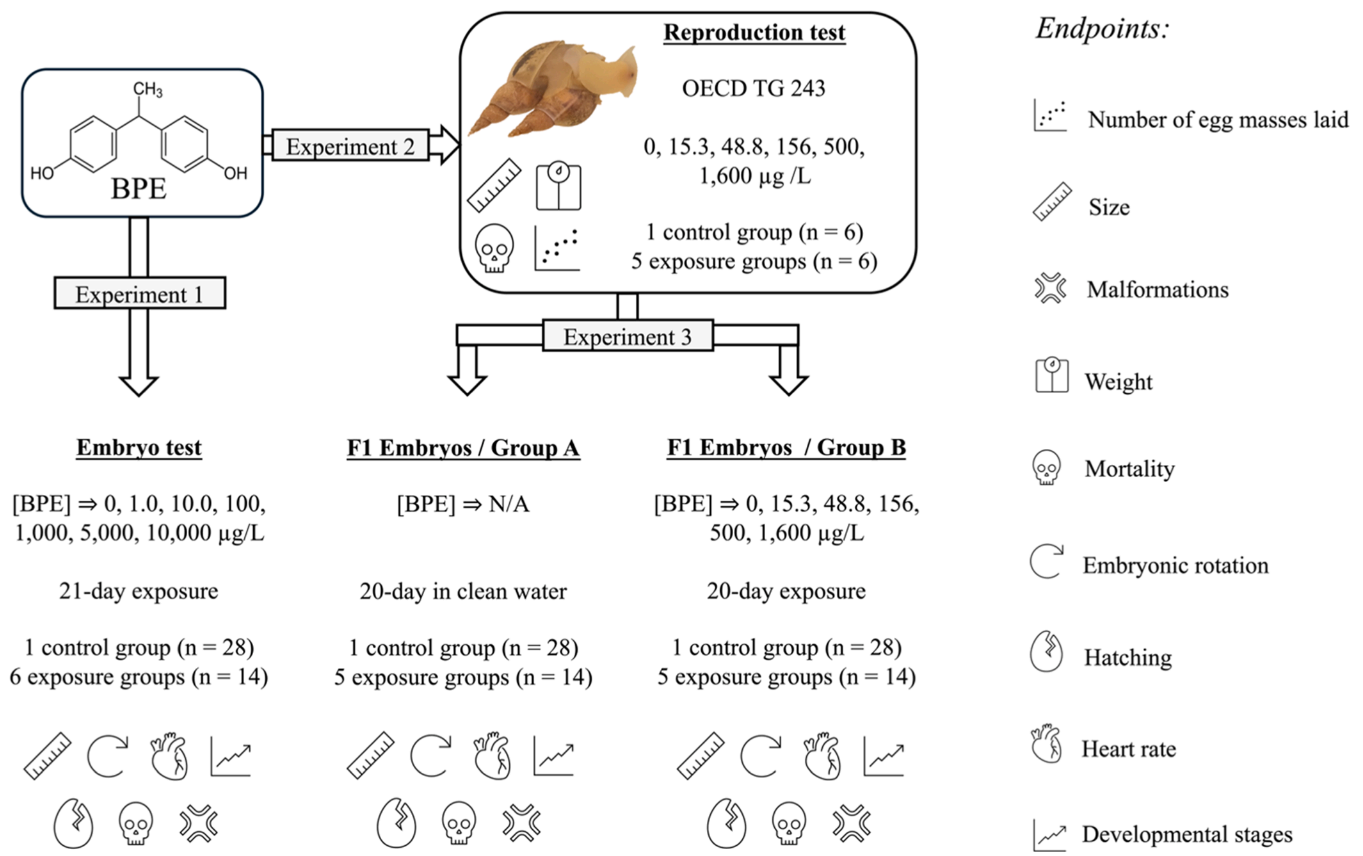

Fig. 1. The experimental design of the two-generation study with nominal concentrations.

(bpm). All videos were taken using an Axiocam 208 colour camera (Buch & Holm A/S, Herlev, Denmark). The developmental stage (morula, trochophora, veliger, and hippo) of each embryo was recorded daily until all embryos had reached the hippo stage. Malformations in the shape of the shell, the body, or coloration of the embryos, as described by Das and Khangarot (2011), were also recorded daily (Das and Khangarot, 2011). The presence of edema was also counted as a malformation. The hatching and mortality rates were evaluated at the end of the test.

2.5. Experiment 2: Reproduction test

Following the OECD TG 243, 180 adult snails with a shell length of 2.5 – 3.0 cm were selected from our acclimatized laboratory colony. Each snail was measured with a caliper (from apex to basal lip) and weighed dry on a precision scale (1.39 ± 0.25 g, $d = 0.001$ g) (Supplementary Materials, Table S1). Snails were introduced into 1 L glass jars in groups of 5, with tap water and constant aeration provided through glass pipettes. The jars were placed at equal distances in a room with a steady temperature of 20 °C and a 16-hour light/8-hour dark cycle. An acclimatization period was observed until reproduction had occurred in all jars before the test began. Snails were fed ad libitum with organic lettuce during the acclimation and test periods, and the jars were thoroughly cleaned once a week by removing the snails and rinsing with tap water.

After acclimation, the jars were randomly assigned to one of the test groups: 0 (control), 15.3, 48.8, 156, 500, and 1600 μg/L BPE. An illustration of the placement of the jars is available in Supplementary Materials (Fig. S1). These concentrations were chosen after an internal range-finding exposure study lasting 14 days (data not published). The final setup consisted of six replicates per concentration, each with five snails for all groups ($n = 30$).

The test lasted 28 days. The water and test media were renewed every second day. Fresh stock solutions were prepared twice a week and stored in the test room for one day before use to avoid temperature shock during test media renewal. The tanks were checked daily, and the number of laid egg masses was reported in every replicate. Dead snails were counted and immediately removed. Physical-chemical parameters (temperature, conductivity, O_2 saturation, water hardness, and pH) were measured twice a week in at least one test medium of each replicate group. Water hardness was also checked in all the control tanks on day 0 and day 28.

At the end of the test, the snails were measured and weighed again, then euthanized in 2.5 % $MgCl_2$ for 24 h.

2.6. Experiment 3: Evaluation of the intergenerational effects of BPE

This study, specifically Experiment 3 (Fig. 1), was conducted as an extension of Experiment 2 (Fig. 1), which was based on the OECD TG 243 reproduction test. This test aimed to assess the impact of parental exposure to BPE on the development of the offspring. F1 generation embryos from the exposed parents were separated into two groups: one developing in clean water and the other developing under the same exposure conditions as the parent snails.

Three egg masses were selected from each concentration group on day 16 and day 17 of Experiment 2 (Fig. 1) and divided into two groups. The first group of eggs, designated as group A, collected on day 16, was placed in clean water to investigate whether the effects observed in directly exposed embryos, i.e., Experiment 1 (Fig. 1), would persist through an eventual parental transfer. The second group of eggs, group B, collected on day 17, were developing in the same nominal BPE concentrations as the parent groups they were selected from: 0 (control), 15.3, 48.8, 156, 500, and 1600 μg/L, to investigate a potential change in the sensitivity to BPE of embryos from exposed parents, compared to embryos directly exposed in Experiment 1 (Fig. 1). For both scenarios, eggs in group A (water) or B (BPE), the same protocol as for the exposed embryo test (Experiment 1) was used, as detailed above. The endpoints were the length at days 3, 6, and 9, the embryonic rotation at day 4, the heart rate at day 6, malformations, hatching, and mortality. The test was stopped when hatching in the surviving controls reached 80 %.

2.7. Statistical analysis

All statistics were performed in R (version 4.5.0). The normality of the data distribution for the length at day 9, rotation, and heart rate of embryos was checked using the Shapiro-Wilk test. For normally distributed data, variance equality was checked with Levene's test, followed by a one-way ANOVA. A post-hoc test was then applied to the data sets to identify the groups with significant differences from the controls. Kruskal-Wallis's test, followed by Dunn's post-hoc test, was applied to data that was not normally distributed. The data about the mortality of the embryos and the adults were checked with Kaplan-Meier curves and a log-rank test. The significance criterion chosen was $p < 0{,}05$. All figures were created using the ggplot2 package in R (version 4.5.0). Boxplots were generated with boxes representing the first quartile (Q1, 25th percentile), the median (Q2, 50th percentile), and the third quartile (Q3, 75th percentile). Whiskers were drawn at Q1 or Q3 ± 1.5 × (Q3 – Q1). Values falling outside the whiskers that were attributable to procedural errors (e.g., eggs damaged during handling) were excluded, whereas values reflecting biological variation were retained. Results are expressed as an average ± standard deviation for the length and the heart rate of the embryos.

3. Results

3.1. Stability of the BPE concentration

LC-MS/MS analyses of the exposure media for the embryo test (Experiment 1) and the intergenerational study (Experiment 3) revealed a better stability of BPE compared to the reproduction test (Experiment 2). In that experiment, BPE showed a sharp decrease in the initial concentration in the test vessels between media renewals. Time-weighted average (TWA) concentrations are reported in Table 1 and will be used over the nominal concentration to report results. More extensive data are available in the Supplementary Materials (Table S2).

3.2. Experiment 1: Abnormal embryonic development at high BPE concentrations

No statistically significant effects were observed in the embryos in any exposed groups up to 725 μg/L, compared to controls, for the endpoints: length, rotation at day 4, heart rate at day 6, developmental stages, hatching, and mortality (Fig. 2). At day 21, the hatching rate was 100 % in exposed groups up to 725 μg/L, 86 % in 5151 μg/L, and 0 % in 11.946 μg/L. Survival analysis revealed significant differences in the timing of hatching across the groups. Embryos in the control group and those exposed to BPE up to 5151 μg/L showed high hatching rates, while embryos exposed to 11,946 μg/L exhibited complete hatching inhibition, with no embryos hatching by day 21 ($\chi^2 = 71.9$, $p = 2.0.10^{-13}$). The overall mortality rate at day 21 was 0 % in all groups, as even malformed and unhatched embryos displayed crucial developmental features, such as a beating heart.

All the embryos grew following an exponential trend, and control embryos reached 1443 ± 153 μm by day 9 (Fig. 2A). The growth of embryos exposed to 5151 and 11,946 μg/L of BPE was significantly affected by BPE, as these embryos were respectively 1.20 (1205 ± 89 μm) and 1.50 (964 ± 77 μm) times smaller than the control embryos on the same day. On day 4, all embryos in all exposed groups were rotating, and no malformation was visually identifiable. The embryonic rotation was significantly lower only in eggs exposed to the highest concentration (11,946 μg/L), with 1.19 rpm compared to 1.60 rpm for the control group (Fig. 2B).

On day 6, a heart rate was visible and quantifiable in at least 90 % of the embryos exposed from 0 (control) to 725 μg/L BPE. One egg was excluded from the statistics in the group exposed to 1.1 μg/L, as it had been damaged during the experimental procedures. Additionally, one embryo in this group didn't have a countable heart rate. At concentrations above 725 μg/L, the proportion of countable beats decreased to 50 % and 0 % of the embryos exposed to 5151 and 11,946 μg/L, respectively. Although no obvious malformation was visible at this time, the embryos displaying a heart rate in group 5151 μg/L, were quantified at 40 ± 13 bpm, less than half the rate observed in control embryos, at 84 ± 13 bpm (Fig. 2C). Heart rates were measured at 76, 80, 78, and 79 bpm in embryos exposed to 6.6, 95.9, 725, and 5151 μg/L BPE, respectively.

At the two highest concentrations, a delay in the general development was also observed with > 90 % of the embryos reaching the hippo stage (last stage of development) when exposed to 5151 μg/L and 11,946 μg/L BPE respectively at day 7 and day 9, when embryos exposed to 725 μg/L and under had reached this stage already at day 6 (Fig. 2D). Malformations were first visible in the highest concentration (11,946 μg/L) around day 7, characterized by (1) body and shell malformations, (2) a discoloration, and (3) the presence of edema (Fig. 2E–G). Overall, malformations were detected in 14 % of the embryos exposed to 5151 μg/L BPE, while 100 % of those exposed to 11,946 μg/L were malformed. A detailed figure showing the malformations observed on embryos exposed to 11,946 μg/L BPE at day 7 is available in Supplementary Materials (Fig. S2).

3.3. Experiment 2: Reproduction of sexually mature snails was not affected by BPE

All individuals in the control group survived. The survival rate was 90 % in groups 4.0, 11.8, and 47.4 μg/L BPE, 63 % in 187 μg/L, and 73 % in the group exposed to the highest concentration, 873 μg/L. However, in one replicate of the group exposed to 187 μg/L, all five snails were dead by day 25, which highly impacted the survival rate of the group. Excluding this replicate, the survival rate of this group increased to 72

Table 1
Nominal and corresponding time-weighted average (TWA) concentrations of BPE for the embryo test (Experiment 1), the reproduction test (Experiment 2), and F1 generation embryos in BPE (Experiment 3, group B).

Experiment 1 Exposed embryos		Experiment 2 Reproduction test		Experiment 3 F1 generation (group B)	
Nominal concentration (μg/L)	TWA concentration (μg/L)	Nominal concentration (μg/L)	TWA concentration (μg/L)	Nominal concentration (μg/L)	TWA concentration (μg/L)
1.0	1.1	15.3	4.0	15.3	6.6
10.0	6.6	48.8	11.8	48.8	13.4
100	95.9	156	47.4	156	78.7
1000	725	500	187	500	218
5000	5151	1600	873	1600	1037
10,000	11,946				

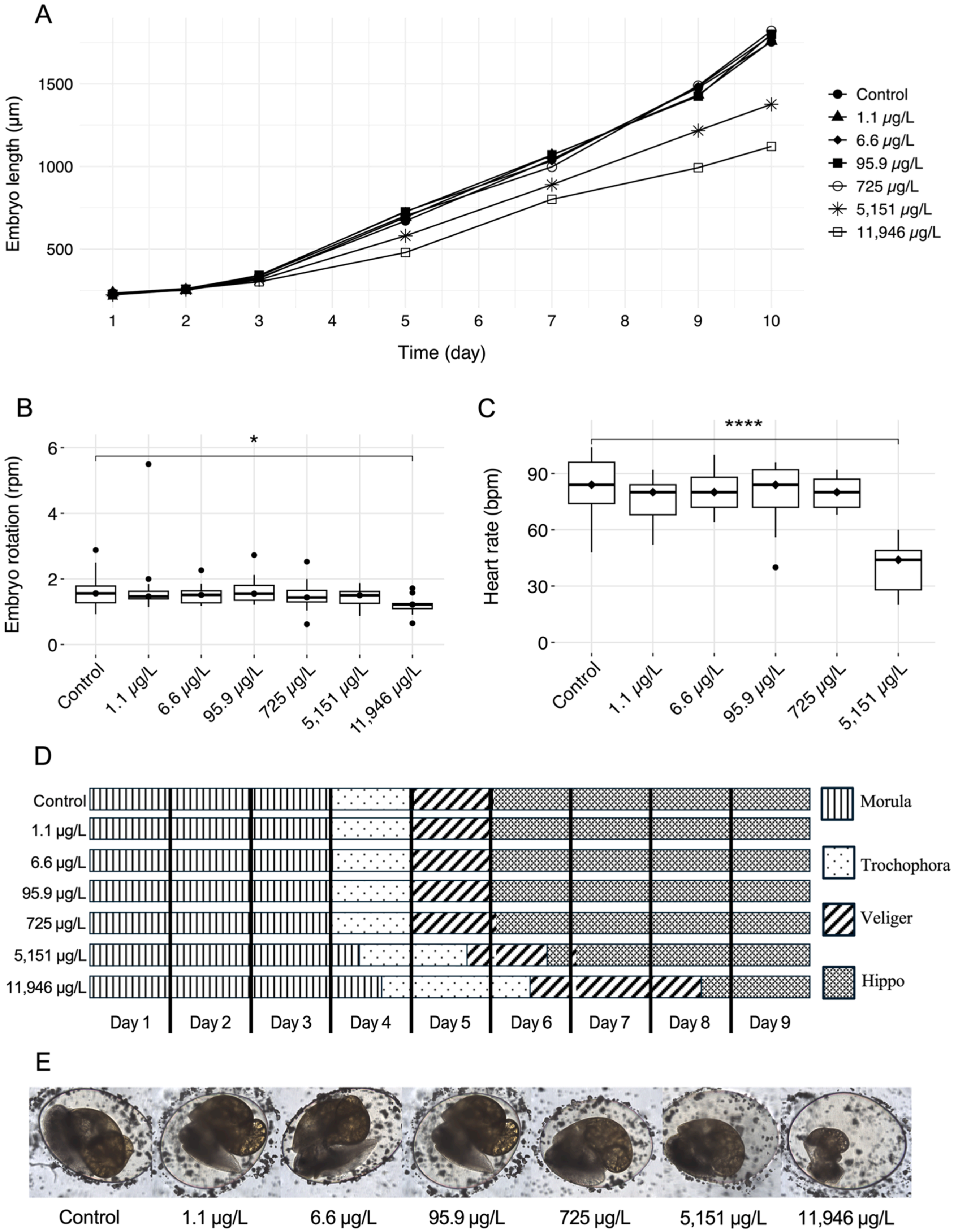

Fig. 2. (A) L. *stagnalis* embryo length (μm) during the first 10 days of a 21-day BPE exposure. (B) Rotation speed (rpm) of L. *stagnalis* embryos at day 4 of BPE exposure. (C) Snails (L. *stagnalis*) embryos' heart rate (bpm) at day 6 of a BPE exposure. Asterisks indicate significant differences from the control (* = $p < 0.05$, **** = $p < 0.0001$) using the Kruskal-Wallis test, followed by Dunn's post hoc test. (D) Evolution of the developmental stages of L. *stagnalis* embryos during the first 9 days of a 21-day BPE exposure test. (E) Representative images of L. *stagnalis* embryos after a 9-day exposure to BPE.

%.

All individuals in all groups experienced growth and weight gain, but there was no statistically significant difference in growth and weight gain between the control and exposed groups. Despite higher mortality in the highest two exposed groups, no statistically significant difference was observed in the number of laid egg masses per individual (Supplementary Materials, Fig. S3). At the end of the test, the LOEC was 187 μg/L BPE, as the log-rank test applied to the Kaplan–Meier survival curves showed a significantly lower survival rate compared to the control ($p = 0.0039$, Fig. S3). The measured physical-chemical parameters of the exposure media are also available in Supplementary Materials (Table S3).

3.4. Experiment 3: A multigenerational approach reveals an increasing sensitivity of F1 generation embryos to BPE

On day 4, all embryos in both groups, A and B, did not have significantly different embryonic rotation speeds compared to their control groups (Fig. 3A and B). At day 6, only the embryos exposed to 1037 μg/L in group A exhibited a significantly lower heart rate compared to the controls (Fig. 3C). In group B, none of the embryos displayed a heart rate significantly different from the controls, measured at 74 bpm (Fig. 3D). On day 9, all embryos in group A (water) reached a length similar to that of the control embryos (860 ± 107 μm) (Fig. 3E). In group B (BPE), after the same time, control embryos grew to a diameter of up to 961 ± 81 μm, while embryos exposed to 218 and 1037 μg/L BPE were significantly smaller, measured at 778 ± 128 and 852 ± 89 μm, respectively (Fig. 3F). At the end of the test, in group A, a total of 1, 2, 2, 1, and 1 embryos were reported as malformed in parental concentration groups 0 (control), 4.0, 11.8, 187, and 873 μg/L, respectively. In group B, in parental concentration groups 4.0, 187, and 873 μg/L, we reported a total of 3, 8, and 1 malformed embryos. Notably, in group B, >50 % of the surviving embryos from parents exposed to 187 μg/L were malformed.

Mortality in group A was 20.00 %, 14.29 %, 21.43 %, 7.14 %, 0.00 %, and 8.33 % for embryos originating from parents exposed to 0 (control), 4.0, 11.8, 47.4, 187, and 873 μg/L BPE, respectively. In group B, mortality was > 20 % for embryos from parents exposed to 4.0 and 187 μg/L (28.57 and 61.54 %, respectively). None of the embryos died when exposed to 0 (control), 13.4, and 78.7 μg/L BPE, but 15.4 % died in the egg group exposed to 1037 μg/L BPE.

The hatching rate of surviving embryos followed the same pattern as the mortality, with the same concentration groups affected by >20 % in groups A and B (Fig. 3G and H). The lowest hatching rate was observed in eggs exposed to 218 μg/L in group B, where only 38.5 % of the surviving embryos hatched on day 19. Interestingly, within the same group, all exposed embryos exhibited a one- to two-day delay in hatching compared to the controls (Fig. 3H).

4. Discussion

4.1. Developmental toxicity of BPE in L. stagnalis embryos

We hypothesized that BPE would negatively impact the embryonic development of L. *stagnalis*. Our observations confirmed this hypothesis, as we observed, among other things, a decrease in embryo length, a decreased heart rate, a delay in embryonic development, and shell and body malformations. Previous studies have reported similar effects of bisphenols on other invertebrate species. For example, the embryogenesis success of the bivalve *Mytilus galloprovincialis* was decreased by exposure to BPA ($EC_{50} = 47.6$ μg/L) (Arslan, 2025). The metamorphosis of the embryos of the mussel *Mytilus coruscus* was also impaired by BPA at concentrations as low as 10.0 μg/L (Huang et al., 2025). Pigmentation impairments were observed in the marine ascidian *Phallusia mammillata*, where exposure to BPE, BPA, and bisphenol F disrupted the differentiation of pigmented cells (Gomes et al., 2019). In the apple snail *Marisa cornuarieti*, BPA caused hormonal alterations that resulted in a decrease in heart rate at concentrations as low as 100 μg/L (Schirling et al., 2006). BPA was also found to affect pigmentation in the embryos of vertebrate species, such as the zebrafish *Danio rerio,* with 25 % of embryos affected at 5000 μg/L, and 100 % from 15,000 μg/L, likely due to endocrine-related mechanisms (Plahuta et al., 2014; Gomes et al., 2019). Given the role of endocrine signaling in regulating the developmental processes affected by bisphenols in invertebrates, we initially investigated BPE as a potential endocrine disruptor affecting the development of L. *stagnalis* embryos.

Both the embryo length and the heart rate were affected at the same concentrations, where a general developmental delay also occurred. An ANCOVA on these two endpoints to assess the impact of either the treatment (BPE) or the delay in the development revealed that both the length and the heart rate are directly affected by the treatment ($p = 7.67 \times 10^{-8}$ and $p = 4.61 \times 10^{-10}$, respectively). However, only the length is shown to be directly affected by the delay of the embryonic development ($p = 3.89 \times 10^{-3}$).

Also, in our study, these effects occurred at very high BPE concentrations, above 5000 μg/L. These concentrations are 50 to 100 times higher than those shown to influence the heart rate in embryos of other snail species exposed to different bisphenols, and more than twice the concentrations that cause impairments in cell pigmentation differentiation in ascidians, for instance (Schirling et al., 2006; Gomes et al., 2019). Lastly, the effects we observed are not exclusive to bisphenols; they also occur when freshwater snails are exposed to other classes of chemicals, e.g., metals, metal nanoparticles, or nonylphenols (Schirling et al., 2006; Lalah et al., 2007; Das and Khangarot, 2011). A delay in reaching consecutive developmental stages and organ development in L. *stagnalis* was induced by nonylphenol at a concentration of 105 μg/L (Lalah et al., 2007). A decrease in the embryonic rotation speed was reported in embryos of the freshwater snail *Indoplanorbis exustus* exposed to copper nanoparticles (Cu-NPs) (Khangarot and Das, 2010; Mehraj and Pandit, 2022). Cu-NPs at 1930 μg/L also caused an alteration in cell pigmentation in the freshwater snail *Indoplanorbis exustus*, compared to cadmium in the freshwater snail *Physa acuta* (Cheung and Lam, 1998; Mehraj and Pandit, 2022).

Considering the low sensitivity of L. *stagnalis* embryos to BPE, along with the range of chemicals with different modes of action shown to produce similar effects, we suggest that BPE does not have specific endocrine-disrupting activity during the development of L. *stagnalis* embryos. Moreover, the observed effects on rotation, heart rate, and hatching are most likely explained by developmental delays in the two highest exposure groups. This caused these endpoints to be evaluated at different stages of the development of the embryos. Instead, we propose that the effects we observed are likely a result of high-dose systemic toxicity. This explanation aligns with previous studies on aquatic invertebrates, although those studies focused on adult individuals rather than embryos. They discussed non-endocrine specific toxicity mechanisms, such as oxidative stress and inflammatory responses, related to other bisphenols like BPA, bisphenol AF, bisphenol F, and bisphenol S, for instance, in the clam *Ruditapes philippinarum* and crabs *Carcinus aestuarii* (Fabrello et al., 2024a, 2024b).

4.2. BPE and the reproduction of sexually mature L. stagnalis

We hypothesized that BPE would act on the reproduction of L. *stagnalis* in the same way BPA does on other snail species, by increasing the number of egg masses laid. Indeed, after 28 days, the reproduction rate was increased for two freshwater snails, *Biomphalaria alexandria* and *Potamopyrgus antipodarum*, exposed to BPA at concentrations as low as 100 μg/L and 40 μg/L, respectively (Sieratowicz et al., 2011; Habib et al., 2023). BPA also enhanced superfemales in the freshwater snail *Marisa cornuarietis* after 5 and 12 months of exposure to levels down to 1 μg/L (Oehlmann et al., 2000). The quality of the sperm of the freshwater clam *Claetura nilotica* was also negatively affected by BPA, showing this

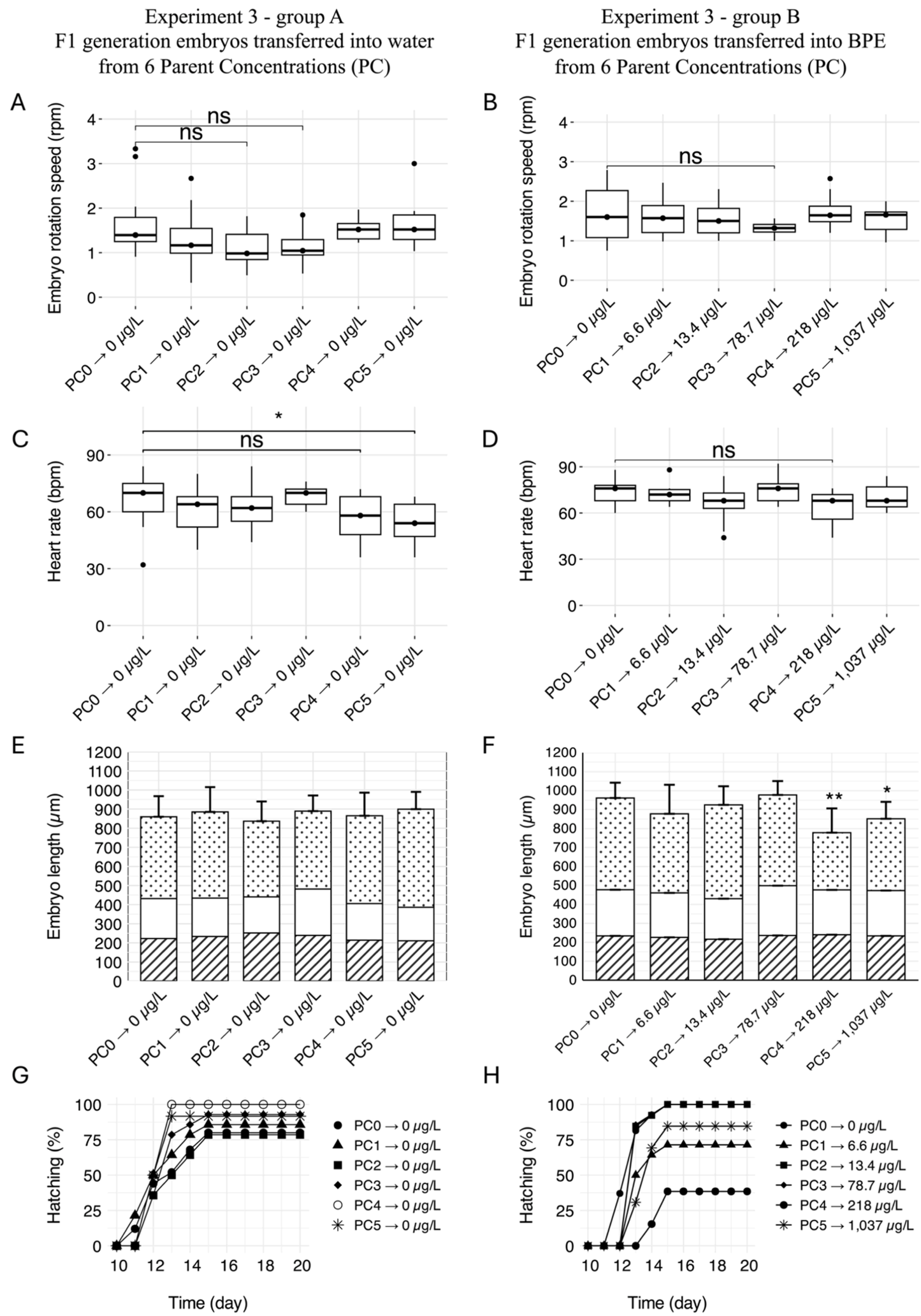

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Fig. 3. Results of the intergenerational study (Experiment 3) for the F1 generation embryos of Group A (A, C, E, G) and Group B (B, D, F, H). (A) Average embryo rotation speed (rpm) of F1 generation embryos in clean water. (B) Average embryonic rotation speed (rpm) of F1 generation embryos in BPE. (C) Average heart rate (bpm) of F1 generation embryos in clean water. (D) Average heart rate (bpm) of F1 generation embryos in BPE. (E) Average length of F1 generation embryos in clean water, and (F) average length of F1 generation embryos in BPE, where stripes indicate days 0 to 3, blank indicates days 4 to 6, and dots indicate days 7 to 9. (G) Hatching rate of F1 generation embryos in clean water, and (H) hatching rate of F1 generation embryos in BPE. Parent Concentrations (PC) for BPE are written as: PC0 = control parents group, PC1 = parents exposed to 4.0 μg/L, PC2 = parents exposed to 11.8 μg/L, PC3 = parents exposed to 47.4 μg/L, PC4 = parents exposed to 187 μg/L, and PC5 = parents exposed to 873 μg/L. Statistical significance is reported using Levene's test followed by a one-way ANOVA and a post-hoc test (C), and the Kruskal-Wallis test followed by Dunn's post-hoc test (A, B, D, F). Asterisks indicate significant differences from control (ns = not significant, * = $p < 0.05$, ** = $p < 0.01$).

chemical to be both estrogenic and antiandrogenic (Sheir et al., 2020). In our reproduction test, we found no statistically significant effect on the number of egg masses laid, even at concentrations up to 873 μg/L BPE. Our results are nonetheless consistent with a study with BPA by Gilroy et al. (2025), where exposure of the freshwater snail *Planorbella pilsbryi* to 479 μg/L did not change the reproduction rate of adults in a 28-day reproduction test (Gilroy et al., 2025).

As for the embryo test, BPE was primarily tested as a potential endocrine-disrupting chemical in L. *stagnalis*. Although studies suggested that the estrogen receptor, or the estrogen-related receptor, for instance, could be involved in endocrine-disrupting mechanisms in *Gastropoda*, BPE is unlikely to exert a direct estrogenic or antiestrogenic effect in this molluscan species (Morales et al., 2018; Scott, 2012, 2013). Indeed, the egg-laying mechanism, investigated here as the primary endpoint, does not involve the binding of either endogenous or exogenous estrogens to the estrogen receptor (Scott, 2012). The reproduction mechanism is controlled by neurohormones produced in cerebral glands like the caudodorsal cells (CDCs) (Wood et al., 2021). The exact mechanism involved in the reproductive axis involving neurohormones in molluscs has yet to be fully described, as well as the toxic mechanisms behind changes in reproduction after BPA exposure, for instance (Ketata et al., 2008; Scott, 2012).

Overall, we did not observe a significant change in the reproduction of sexually mature L. *stagnalis* individuals exposed to BPE. The decreasing number we identified in the number of laid egg masses, although not significant, is likely due to changes in energy allocation in response to exposure to high concentrations of xenobiotics, where snails prioritize survival over reproduction (Osborne et al., 2020). Thus, we advance that BPE does not cause sublethal endocrine-related effects on the reproduction of sexually mature L. *stagnalis*.

If significant impacts are observed on endocrine-specific endpoints in mollusc species, such as the number of egg masses, it is essential to investigate the underlying mechanisms. This can be done through histopathological analysis to examine potential malformations in the central nervous system, such as in CDCs. Beyond the qPCR analysis of selected genes, transcriptomic approaches could be employed to quantify alterations in the expression of neurohormones regulating the reproductive system of L. *stagnalis*. Moreover, integrative multi-omics strategies (e.g., combining proteomics, transcriptomics, metabolomics, and lipidomics) would provide a systems-level perspective on the biochemical and molecular pathways perturbed by exposure, thereby facilitating mechanistic links between observed reproductive and developmental effects and underlying molecular processes.

4.3. The impacts of parental exposure on the fitness and the sensitivity of F1 generation offspring

The intergenerational study evaluated BPE toxicity in F1 generation embryos with two distinct approaches. Embryos in group A (water) assessed the potential parental toxicity through the transfer of toxicants or metabolites (e.g., via gametes or egg provisioning) or by inducing increased sensitivity in the embryos. Embryos in group B (BPE) were used to assess the change in sensitivity of F1 generation embryos to BPE resulting from epigenetic modifications or alterations in hormonal signaling pathways, as compared to directly exposed embryos from unexposed parents.

Our findings do not support our hypothesis of a parental toxicity transfer, as no significant effect was observed in any endpoint, except for the heart rate in embryos from the highest-exposed parent group (873 μg/L). In Experiment 1, a decreased heart rate was associated with developmental delays, likely due to the high exposure concentrations used in the test, which were not achieved in our intergenerational study. An explanation for the observed decrease in heart rate may lie in the link between cardiac activity and physiological stress (Capela et al., 2024). Transferring eggs from a contaminated environment to clean water could disrupt their general homeostasis, potentially leading to reduced heart rates in the embryos. However, since the transfer occurred at the morula stage, prior to heart formation, the relevance of osmotic imbalance as a direct cause of altered heart rate is uncertain. Early stress exposure may affect later cardiac development indirectly, but further investigation is needed to confirm this mechanism.

In group B, embryos developing at the same nominal BPE concentrations as their parents showed no significant changes in embryonic rotation speed or heart rate. These results are in agreement with the results from Experiment 1 and the F1 generation embryos in group A. Nevertheless, although external parameters like oxygen levels, for instance, can affect embryonic rotation, this endpoint is not used in the latest exposure tests on embryos of L. *stagnalis* (Capela et al., 2024). This endpoint was considered not sensitive enough to assess the effects of EDCs in this snail species due to its early measurement time in embryonic development (day 4 – trochophore stage). However, the observation of similar effects in embryos in group B from Experiment 3 and embryos from Experiment 1 for the other endpoints, e.i. embryo length, hatching rate, malformations, and mortality, but at lower concentrations, shows that F1 generation L. *stagnalis* embryos are more sensitive to BPE when derived from exposed parents.

To our knowledge, this is the first time BPE has been tested on L. *stagnalis*. While it is the first time such an intergenerational study has been performed on L. *stagnalis* for the risk assessment of this BPA alternative, it is not the first time a multigenerational approach has been carried out on molluscs, though more frequently for pharmaceuticals, pesticides, and metals (Bal et al., 2017; Harayashiki et al., 2024). Our results suggest that parental exposure indeed leads to the production of embryos more sensitive to BPE exposure than those from unexposed parents, as the lowest observed effect concentration (LOEC) for embryonic growth at 218 μg/L in our intergenerational study was >20-fold lower than that for embryos from unexposed parents (5151 μg/L BPE). In a study with BPA and its alternatives in the freshwater snail *Planorbella pilsbryi*, F1 embryos from exposed parents were 100-fold more sensitive to BPA exposure than embryos from non-exposed parents (Gilroy et al., 2025). This also aligns with studies on other snail species and other substances, where long-term exposure to, for instance, psychoactive substances, EDCs, or steroids, could be a threat to an entire population by increasing sensitivity across generations (Bal et al., 2017; Borysko and Ross, 2014; Henry et al., 2022). Our LOEC for F1 generation embryos from exposed parents remains 100 times higher than the reported BPE concentrations in freshwaters in Europe (2.77 μg/L), but lower sensitivity may appear in future generations of exposed embryos (F2, F3, etc.) (Reátegui-Zirena et al., 2017; Russo et al., 2021).

4.4. Limitations and environmental implications

Our study is the first to evaluate the developmental effects of BPE on embryos of a mollusc species, both through direct and indirect exposures, and in an intergenerational study, combined with an adult reproduction test. Our conclusions are limited because we do not provide directly comparable experimental data for BPA in L. *stagnalis*, although such data are available for other freshwater invertebrate species (Mihaich et al., 2009; Cosentino et al., 2022). Our study evaluates endpoints previously described in the literature for both developmental and reproductive toxicity of L. *stagnalis*. However, the toxicity mechanisms have not been investigated, which hinders a thorough understanding of the pathways involved in the observed outcomes. Yet, the toxicity mechanisms are not always comparable, particularly between vertebrate and invertebrate species, which again encourages further investigations of the endocrine system of molluscs. The effects of BPE on embryos appear at lower concentrations in offspring from exposed parents. However, as it was decided to stop investigating the parental toxicity transfer and the sensitivity of the offspring at the F1 generation, we are unable to verify whether the same pattern would be observed in the F2 or F3 generations. A study on the toxicity of diclofenac by Bouly et al. (2022) highlighted differences in the responses of F1 and F2 generations of L. *stagnalis* (Bouly et al., 2022). In that study, food intake and reproduction were lowered for the F1 generation but increased for the F2 generation. The F1 generation adopted an energy conservation strategy by reducing reproduction rates, while the F2 generation responded by compensating with increased fecundity, resulting in a higher number of egg masses per snail. Such response strategies can be adopted at different generation levels and life stages, which our experimental setup does not allow us to investigate. Moreover, our intergenerational study is itself an extension of the OECD TG 243 reproduction test. As the exposure time of the parents could impact the response of the embryos, the experiment was designed to collect all embryos for one experiment on the same day to avoid external variations, which limits the number of available F1 individuals.

Additionally, in the reproduction and the F1 generation embryo tests, the calculated TWA concentrations revealed a gap in the BPE concentration compared to the initially chosen nominal concentrations. Chemical analyses are thus essential for accurately evaluating toxicological parameters, such as the EC_{50} or the LC_{50}, and have comparable data across laboratories. Considering the importance of invertebrate species in aquatic ecosystems and the potential risk of long-term exposure to BPA alternatives, such as BPE, more multigenerational-type studies are necessary to understand the changes in sensitivity across generations. Such approaches will be required to safely assess the toxicity of chemicals, including BPA alternatives, used in the industry. Our findings highlight concerns regarding the long-term exposure of mollusc populations to BPA alternatives in natural environments. This exposure may lead to more sensitive generations, resulting in a decline in the fitness of the population and ultimately decreasing the overall number of individuals.

The transgenerational effects of BPE are not the only concern regarding BPA alternatives. As outlined in the introduction, BPE is used alongside 20 other analogues in commercial products (Yang and Yu, 2025). Consequently, these compounds are released into the environment together, making single-compound exposures highly unlikely under natural conditions. Therefore, risk assessment efforts for BPA alternatives should account for mixed exposures in order to investigate potential additive or synergistic effects among these bisphenols, which may further contribute to transgenerational impacts.

Synopsis

Data on the toxicity of bisphenol A alternatives in aquatic invertebrates remain limited. This study investigates the toxicity of bisphenol E on two generations of *Lymnaea stagnalis*.

CRediT authorship contribution statement

Gaëtan Y. Tucoo: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jane E. Morthorst:** Writing – review & editing, Supervision. **Elvis Genbo Xu:** Writing – review & editing, Supervision. **Henrik Holbech:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.aquatox.2025.107641.

Data availability

Data will be made available on request.

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